LDS Standard Data format

What is Cycle Counting?

Cycle counting transforms a complex load-time history into a simplified representation of stress cycles used for fatigue analysis.

As illustrated in the figure below, the continuous signal (top) is decomposed into individual cycles characterized by:

- a stress range $\Delta\sigma$,
- a mean stress σ_m,
- a number of repetitions n_i.

Instead of using the full signal, only the equivalent fatigue cycles are retained for damage calculation.

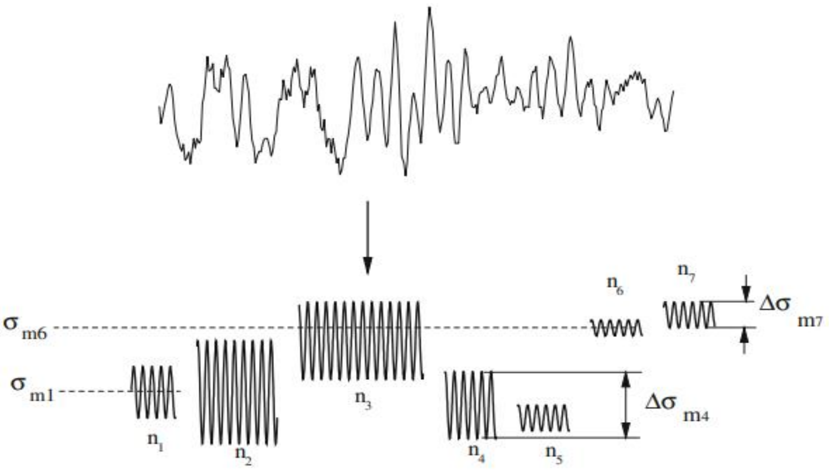


Figure 1 : Schematic representation of the application of a cycle counting method (Vassilopoulos & Keller, 2017).

What is an LDS file?

An LDS (Load Data Series) file contains the sequence of stress extrema (peaks and valleys) extracted from a load-time history. **The file must contain successive turning points only.**

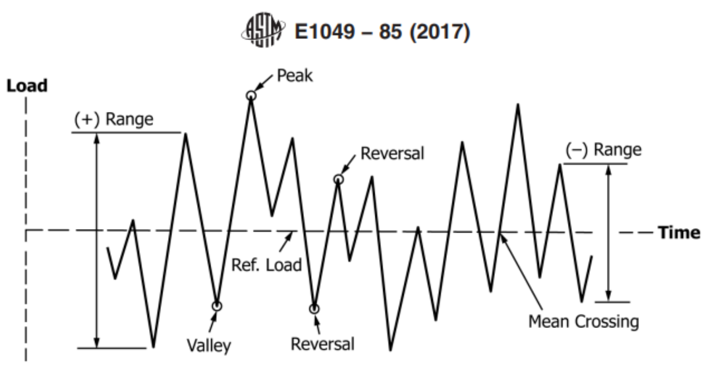


Figure 2 : Basic Fatigue Loading Parameters (ASTM E1049-85, 2017).

File naming convention

File: LDS_{ResearcherLastname}_{YYYY-MM}.csv

- YYYY-MM = starting month of the experiment
- Example: LDS_Dupont_2024-03.csv

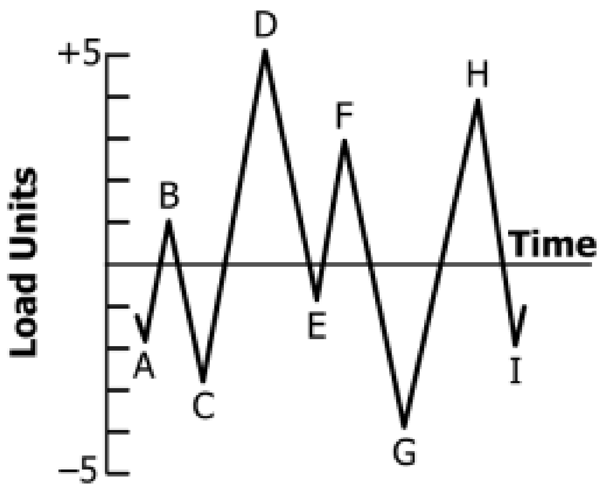
File format requirements

- Encoding: UTF-8
- Separator: comma (,)
- File type: .csv
- Header required
- Units: MPa

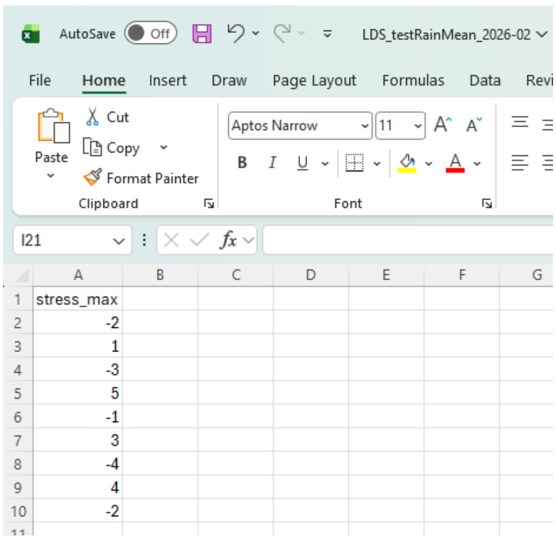
Required column (column names must be exact)

Variable name	Description	Symbol	Unit	Data type	Mandatory
stress_max	Max stress	sigma_max	[MPa]	double	y

Example of valid LDS file



Original load history



LDS file

For the CSV file, Microsoft Excel works well for editing and saving the data.
However, other tools such as LibreOffice Calc, Google Sheets, or any text editor (e.g., Notepad++, VS Code) can also be used, provided the file is saved in CSV format (UTF-8 encoding).